

分析化学第二次作业

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1.写出IUPAC的英文全称，说明灵敏度和检测限的定义和相互关系

- IUPAC的全称是International Union of Pure and Applied Chemistry，即国际纯粹与应用化学联合会。
- 灵敏度：灵敏度是描述分析方法对被分析物浓度或质量变化的响应能力的指标。定量而言，是工作曲线的斜率。
- 检测限：信号强度等于噪音标准偏差的3倍时可检测的样品浓度称为检测限。
- 两者的关系：检测限 $= \frac{3S_b}{S}$ ，其中 $S_b = \sqrt{\frac{\sum_{i=1}^N (N_i - \bar{N})^2}{n-1}}$ ， S 为灵敏度

2.

3. 实验室中有四种天平，其性能见下表

	台 称	普通天平	分析天平	半微量天平
最大载重	100g	200g	200g	20g
感量(分度值)	0.1g	1mg	0.1mg	0.01mg

为下列称量选择合适的天平：

- (1) 准确称取基准邻苯二甲酸氢钾约 0.5g，以标定 NaOH 溶液的浓度；
- (2) 称取 10g 工业 $K_2Cr_2O_7$ ，配取铬酸洗涤液；
- (3) 称取甲基橙，配制 0.1% 甲基橙溶液 100mL；
- (4) 称一块约 4g 重的铂片，要求准确到小数点后第五位。

(1)

因为只用0.5g，而且需要准确称取基准物质，含量标准要大于99.9%，也即 $0.5g \times 1000 \times 99.9\% = 499.5mg$ 。要精确到0.1mg，故使用分析天平最合适。

(2)

称取10g工业 $K_2Cr_2O_7$ ，即工业纯，含量标准大致在95%上即可，即 $10g \times 95\% = 9.5g$ ，因此使用台秤即可。

(3)

配制甲基橙溶液，甲基橙溶液浓度较低，可近似认为溶液密度与水相同，因此需要 $100\text{mL} \times 0.1\% \times 1\text{g/mL} = 0.1\text{g}$ 左右的甲基橙，含量标准大于99%，故 $0.1\text{g} \times 99\% \times 1000 = 99\text{mg}$ ，故使用普通天平是比较合理的。

(4)

要求精确到小数点后5位，那么就是0.01mg的精度，因此使用半微量天平最合适的。

3.

Blood plasma proteins of patients with malignant breast tumors differ from proteins of healthy people in their solubility in the presence of various polymers. When the polymers dextran and poly(ethylene glycol) are mixed with water, a two-phase mixture is formed. When plasma proteins of tumor patients are added, the distribution of proteins between the two phases is different from that of plasma proteins of a healthy person. The distribution coefficient (K) for any substance is defined as $K = [\text{concentration of the substance in phase A}]/[\text{concentration of the substance in phase B}]$. Proteins of healthy people have a mean distribution coefficient of 0.75 with a standard deviation of 0.07. For the proteins of people with cancer, the mean is 0.92 with a standard deviation of 0.11.

(a) Suppose that K were used as a diagnostic tool and that a positive indication of cancer is taken as $K \geq 0.92$. What fraction of people with tumors would have a false negative indication of cancer because $K < 0.92$?

(b) What fraction of healthy people would have a false positive indication of cancer? This number is the fraction of healthy people with $K \geq 0.92$, shown by the shaded area in the graph below. Estimate an answer with Table 4-1

Distribution coefficients of plasma proteins from healthy people and from people with malignant breast tumors. [Data from B. Y. Zaslavsky, "Bioanalytical Applications of Partitioning in Aqueous Polymer Two-Phase Systems," *Anal. Chem.* **1992**, 64, 765A]

Table 4-1 见下页

TABLE 4-1 Ordinate and area for the normal (Gaussian) error curve, $y = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$								
$ z ^a$	y	Area ^b	$ z $	y	Area	$ z $	y	Area
0.0	0.398 9	0.000 0	1.4	0.149 7	0.419 2	2.8	0.007 9	0.497 4
0.1	0.397 0	0.039 8	1.5	0.129 5	0.433 2	2.9	0.006 0	0.498 1
0.2	0.391 0	0.079 3	1.6	0.110 9	0.445 2	3.0	0.004 4	0.498 650
0.3	0.381 4	0.117 9	1.7	0.094 1	0.455 4	3.1	0.003 3	0.499 032
0.4	0.368 3	0.155 4	1.8	0.079 0	0.464 1	3.2	0.002 4	0.499 313
0.5	0.352 1	0.191 5	1.9	0.065 6	0.471 3	3.3	0.001 7	0.499 517
0.6	0.333 2	0.225 8	2.0	0.054 0	0.477 3	3.4	0.001 2	0.499 663
0.7	0.312 3	0.258 0	2.1	0.044 0	0.482 1	3.5	0.000 9	0.499 767
0.8	0.289 7	0.288 1	2.2	0.035 5	0.486 1	3.6	0.000 6	0.499 841
0.9	0.266 1	0.315 9	2.3	0.028 3	0.489 3	3.7	0.000 4	0.499 904
1.0	0.242 0	0.341 3	2.4	0.022 4	0.491 8	3.8	0.000 3	0.499 928
1.1	0.217 9	0.364 3	2.5	0.017 5	0.493 8	3.9	0.000 2	0.499 952
1.2	0.194 2	0.384 9	2.6	0.013 6	0.495 3	4.0	0.000 1	0.499 968
1.3	0.171 4	0.403 2	2.7	0.010 4	0.496 5	∞	0	0.5

a. $z = (x - \mu)/\sigma$
b. The area refers to the area between $z = 0$ and $z =$ the value in the table. Thus the area from $z = 0$ to $z = 1.4$ is 0.419 2. The area from $z = -0.7$ to $z = 0$ is the same as from $z = 0$ to $z = 0.7$. The area from $z = -0.5$ to $z = +0.3$ is $(0.191\ 5 + 0.117\ 9) = 0.309\ 4$. The total area between $z = -\infty$ and $z = +\infty$ is unity.

(a)

由题可知，癌症患者的平均分配系数就是0.92，因此被错误判别成假阴性的概率是50%。

(b)

由题可知，正常患者的平均分配系数是0.75，方差为0.07，我们要计算正态分布概率大于0.92的部分，先对其做归一化 $\frac{0.92-0.75}{\sqrt{0.07}} = 2.43$ ，查表可得面积大约是0.4918，故假阳性的概率为 $50\% - 49.18\% = 0.82\%$ 。